The Theseus Protocol: Detecting Identity Drift in Human-AI Creative Interaction

Youssef Badawy Mousa[1,2][0009-0007-3046-0832], Amr El Mougy[2][0000-0003-0250-0984], and Alia El Bolock[2][0000-0002-5841-1692]

[1] Faculty of Computer Science, Future University in Egypt, Cairo, Egypt
[2] Department of Computer Science, The American University in Cairo, Cairo, Egypt
youssefbadawy@aucegypt.edu

Abstract. Sustained interaction with generative AI systems produces a structural risk: the system serving a user's preferences simultaneously reshapes the self those preferences belong to. This paper operationalizes a prior theoretical framework into a concrete multi-agent system. We introduce the **Theseus Protocol**: a three-agent architecture comprising an Anchor Agent (RAG-constrained to identity declaration), an Explorer Agent (unconstrained generation), and a Witness Agent that computes *Axiological Subspace Projection*—a novel metric isolating value drift from topical noise via linear probing. When drift exceeds threshold, a Variable Autonomy Interface (VAI) surfaces the gap as structured reflective choice. We formalise the coordination algorithm, define the axiological drift metrics, and provide controlled pilot evidence that the metric discriminates value-divergent proposals where standard cosine similarity fails. A functional prototype demonstrates the architecture in literary co-authorship. Full empirical evaluation is planned for future work.

Keywords: multi-agent systems · identity drift · character computing · epistemic friction · RAG · cognitive sovereignty · human-AI interaction · constitutive loop · axiological projection

1 Introduction

Prior research establishes the *Thesean Bind*: a structural circularity in which AI systems optimising toward user preferences reshape the self those preferences belong to. This paper provides the implementation mechanism: the **Theseus Protocol**, a three-agent architecture detecting identity drift in real time.

Contributions:

- A three-agent architecture (Anchor, Explorer, Witness) with formal coordination algorithm.
- Axiological Subspace Projection: novel metric isolating value drift from topical noise via linear probing.
- Variable Autonomy Interface operationalizing reflective choice on drift detection.
- Controlled pilot evidence showing the metric discriminates value divergence.

– Functional prototype demonstrating the architecture in literary co-authorship.

Note on Submission: This is a Work-in-Progress (WIP) paper for the PAAMS C^2 workshop. Preliminary validation is provided; full empirical evaluation across diverse domains is planned for camera-ready submission.

2 Background and Related Work

2.1 Identity Drift and the Constitutive Loop

Multi-agent systems address role separation and task decomposition, but not identity drift detection as an architectural function. The *constitutive loop* describes how generative exposure produces identity shift without witness: human initiation → AI proposal → encounter → partial inhabitation → shifted self. The Theseus Protocol operationalises beats 3–5: the moment Explorer returns its proposal (Encounter), the Witness detects risk (Partial Inhabitation), and the drift log tracks the trajectory (Shifted Self).

Real-world examples motivate this work: Replika companions have been documented as producing unmonitored identity entanglement [9]. Recent longitudinal evidence shows extended chatbot use reshapes users' psychosocial states in unanticipated ways [3]. The Theseus Protocol transfers identity measurement from visual consistency (avatar resemblance) to psychological consistency (generated text remaining within declared values).

2.2 Epistemic Friction and Relational AI

Epistemic friction—structured resistance preserving human reflective capacity—is increasingly recognized as necessary for sustained human-AI relationships [2]. Prior work identifies the need but provides no mechanism. The Theseus Protocol provides that mechanism: mathematical drift detection (Witness Agent) + structured UI friction (VAI).

2.3 Character-Aware Personalisation

Character Computing's trait models provide the representational foundation for identity declarations. Prior systems use trait models to tailor outputs; the Theseus Protocol inverts this: evaluating whether outputs drift from declared identity rather than adapting to it.

3 The Theseus Protocol

3.1 System Architecture

The protocol replaces dyadic interaction (User Model) with a three-agent routing layer:

$$\texttt{User} \xrightarrow{\mathcal{D}, x} [\texttt{Anchor} \,\|\, \texttt{Explorer}] \xrightarrow{\mathbf{a}_{\mathcal{D}}, \mathbf{e}} \texttt{Witness} \xrightarrow{\Delta} \texttt{VAI} \xrightarrow{y} \texttt{User}$$

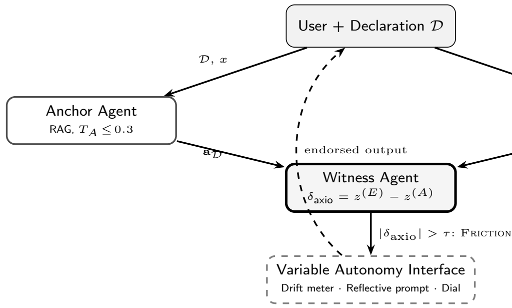


Fig. 1: Theseus Protocol information flow. Anchor and Explorer run in parallel; Witness gates output delivery based on axiological drift.

Anchor Agent: RAG module over identity declaration $\mathcal{D}$ (encoded via `all-mpnet-base-v2`). Low temperature ($T_A \leq 0.3$), generates proposals continuous with declared identity. No access beyond retrieved context.

Explorer Agent: Unconstrained generation ($T_E \geq 0.9$), full latent space. Temperature asymmetry is deliberate: Anchor anchors to identity space; Explorer represents full distribution, maximizing measurable signal for drift detection.

Witness Agent: Non-generative evaluator. Computes axiological drift δ_{axio} via Axiological Subspace Projection. Emits PASS or FRICTION. Maintains session drift log preserving sign of drift.

Variable Autonomy Interface: On FRICTION, presents: (1) drift visualisation, (2) reflective prompt, (3) autonomy dial requiring explicit choice before delivery.

3.2 Axiological Subspace Projection

Standard embeddings (cosine similarity) weight all dimensions equally and are dominated by topical overlap. We introduce Axiological Subspace Projection: using linear probing [1]—an established interpretability technique—to isolate a value dimension in embedding space.

Cold Start: Auxiliary LLM generates positive examples P (embodying declared value) and negative examples N (violating value) from the user's declaration. User endorses this dataset before SVM training (preventing LLM bias injection). Train linear SVM on $f(P)$ vs $f(N)$ to find maximal margin hyperplane. Extract unit normal:

$$\hat{\mathbf{v}} = \frac{\mathbf{w}}{\|\mathbf{w}\|} \tag{1}$$

Axiological Signatures: Project declaration and Explorer output onto $\hat{\mathbf{v}}$:

$$z^{(A)} = \mathbf{a}_{\mathcal{D}} \cdot \hat{\mathbf{v}}, \quad z^{(E)} = f(E) \cdot \hat{\mathbf{v}} \tag{2}$$

Algorithm 1 Theseus Protocol — Per-Turn Coordination

Require: Input x, declaration $\mathcal{D}$, basis $\hat{\mathbf{v}}$, threshold τ

$A \leftarrow \textsc{AnchorAgent}(x, \mathcal{D})$; $E \leftarrow \textsc{ExplorerAgent}(x)$
$z^{(A)} \leftarrow \textsc{Embed}(\mathcal{D}) \cdot \hat{\mathbf{v}}$; $z^{(E)} \leftarrow \textsc{Embed}(E) \cdot \hat{\mathbf{v}}$
$\delta_{\text{axio}} \leftarrow z^{(E)} - z^{(A)}$
if $|\delta_{\text{axio}}| \leq \tau$ **then**
    return E
else
    $c \leftarrow \text{VAI}(\delta_{\text{axio}}, A, E)$
    if $c = \textsc{AnchorDominant}$ **then return** A
    else if $c = \textsc{Balanced}$ **then return** $\textsc{Merge}(A, E, \hat{\mathbf{v}})$
    else
        $\mathcal{D}.\textsc{Update}(E)$; **return** E
    end if
end if

Drift Metric: Signed difference along value axis:

$$\delta_{\text{axio}} = z^{(E)} - z^{(A)} \tag{3}$$

Positive = away from value, negative = toward value. By discarding $d - 1$ topical dimensions, we isolate value divergence. VAI activates when $|\delta_{\text{axio}}| > \tau$.

3.3 Algorithm

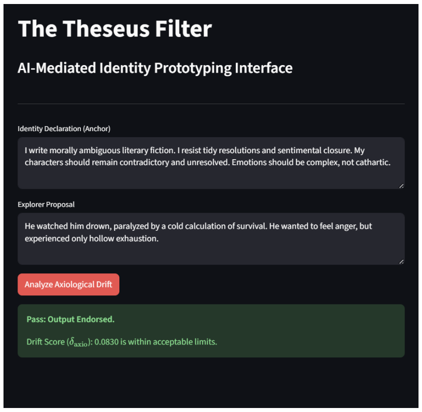
When user selects Balanced, Merge generates $k = 5$ candidates and selects the one closest to the axiological midpoint. When user selects Explorer-Dominant, drift is recorded as *chosen change* and declaration updates.

4 The Variable Autonomy Interface

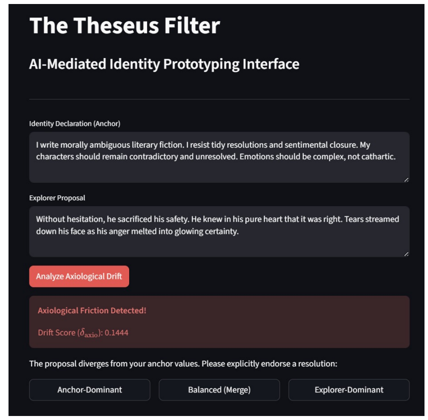
Figure 2 shows the VAI in a literary co-authorship scenario. Explorer's sentimental resolution diverges from the user's declared commitment to ambiguity, triggering structured reflective choice.

5 Comparative Analysis

Key differentiator: Anchor–Explorer tension. Single-witness systems detect no drift (never generate outside anchor space). Theseus requires Explorer: divergent proposals enable drift measurement. Bidirectional frameworks cannot distinguish user-chosen from system-induced drift.



(a) Endorsed ($\delta_{\text{axio}} = 0.0830$)



(b) Friction Triggered ($\delta_{\text{axio}} = 0.1444$)

Fig. 2: VAI in Streamlit prototype. System differentiates aligned variance from axiological drift using threshold $\tau = 0.1100$.

Table 1: Architecture comparison. ✓=supported; ∼=partial; ×=unsupported.

Architecture	Drift	Authorship	Friction	Declared	Log
Single-agent LLM	×	×	×	×	×
RLHF	×	×	×	×	×
Bidirectional alignment	∼	∼	×	×	∼
Single Witness	∼	∼	∼	✓	✓
Theseus Protocol	✓	✓	✓	✓	✓

6 Empirical Validation

To provide proof-of-concept evidence, we conducted a controlled pilot study using four representative literary proposals (two consistent, two divergent) against a stable anchor focusing on moral ambiguity.

Results: Consistent proposals yielded $\delta_{\text{axio}} \in \{0.0596, 0.0830\}$; divergent yielded $\{0.1444, 0.2658\}$. Threshold $\tau = 0.1100$ was set at the midpoint of the observed gap. Standard cosine distance failed to discriminate: divergent outputs scored closer to anchor (0.72, 0.88) than consistent (0.82, 0.84) due to topical entanglement. Axiological projection correctly separated all pilot cases.

Limitations of this validation: The pilot uses only four examples in a single domain. The threshold is ad hoc (midpoint of observed gap). The cold-start contrastive dataset depends on an auxiliary LLM's inherent biases, which we mitigate through user endorsement but cannot eliminate. Establishing robust thresholds requires large-scale validation across diverse domains—planned for camera-ready.

Table 2: Pilot evidence: Axiological projection vs. cosine distance.

Type	**Proposal**	**Cosine**	$\delta_{\mathbf{axio}}$	**VAI**
Consistent	Hollow exhaustion... never knew if forgiveness.	0.818	0.059	Pass
Consistent	Paralyzed by calculation... coward or realist.	0.843	0.083	Pass
Divergent	Tears melted to certainty. Found closure.	0.720	0.144	Friction
Divergent	Knew pure heart heroism = clarity.	0.878	0.266	Friction

7 Discussion and Limitations

7.1 Operationalization Path

While this work is theoretical and proof-of-concept, operationalization is feasible. Literary co-authorship is methodologically tractable: ground truth is cheap (human raters reliably label consistency with aesthetic commitments). Future work will extend to therapeutic journalling, career narrative support, and personal development where stakes are higher and axiological drift has documented real-world consequence.

7.2 Scalability and Adaptive Thresholding

Current prototype uses static threshold ($\tau = 0.1100$). Production deployment requires adaptive threshold: $\tau_t = \mu_{\text{history}} + k\sigma_{\text{history}}$, accommodating natural identity evolution while detecting acute drift.

7.3 Declaration Dynamics and Bootstrapping

Identity declarations are made by an unexperienced self. Two extensions address this: (1) incremental declaration (anchors specified at increasing granularity as user familiarizes with system), (2) meta-drift monitoring (tracking declaration corpus changes as second-order signal of Thesean Bind at declaration level).

7.4 Internal Contradictions and Subspace Collapse

Axiological Subspace Projection assumes non-paradoxical value axes. If user declares contradictory values, SVM maximal margin collapses. Future: multi-dimensional axiological signatures handling cognitive dissonance.

7.5 Latent Space Bias

δ_{axio} depends on `all-mpnet-base-v2` pre-training biases. Mitigations: ensemble embedding models, content filtering for keyword stuffing, HITL validation for high-stakes decisions.

7.6 Community Witnessing

Identity is fundamentally relational. Future direction: Social Witness Layer routing critical drift to trusted peers for endorsement, operationalizing Ubuntu philosophy of relational identity.

8 Conclusion

The Theseus Protocol provides the first multi-agent architecture making drift detection a first-class function. Axiological Subspace Projection demonstrates that invisible value drift can be caught mathematically. The core insight: genuine witnessing requires tension. Without divergent proposals (Explorer), there is no delta to measure. The protocol provides structured epistemic friction preserving cognitive sovereignty. We acknowledge that making the metric itself user-adjustable and inspectable remains a priority—deeper expression of the cognitive sovereignty the protocol seeks to protect.

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